

Unit-3

TOPIC:- Basics of IoT Networking, IoT Components, Functional components of IoT

OBJECTIVES:- The Internet of Things (IoT) is aimed at **enabling the interconnection and integration of the physical world and the cyber space**. It represents the trend of future networking, and leads the third wave of the IT industry revolution.

OUTCOME:- To conclude, IoT has the capability to create public value and the intended public value dimensions have widened from the goal of improved efficiency. There has been a shift in value creation with other goals and outcomes such as **effectiveness, transparency and collaboration gaining increased presence**.

Basics of IoT Networking, IoT Components, Functional components of IoT

An IoT network refers to a collection of interconnected devices that communicate with other devices without the need for human involvement, such as autonomous cars, smart appliances, and wearable tech.

The network infrastructures most associated with IoT networks are 4G LTE and 5G which are built to support the resource demands of the IoT.

IoT Networks: 5G

IoT is a central use case of 5G infrastructure. This is because the current mobile networks are already struggling to keep up with the resource demands of the exploding IoT device market.

All three major usage scenarios for 5G networks as described by the International Telecommunication Union (ITU) support IoT devices to some extent: enhanced mobile broadband (eMBB), massive machine-type communications (mMTC), and ultra-reliable and low-latency communications (URLLC).

Enhanced Mobile Broadband

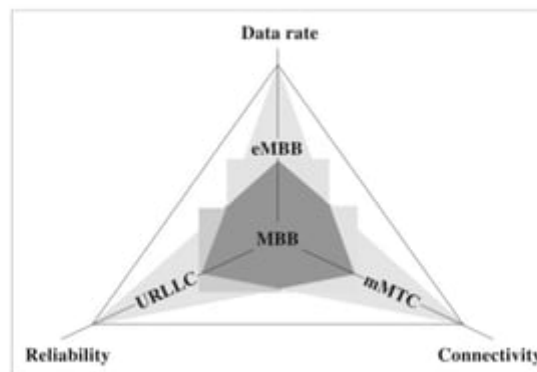
According to Cisco's Visual Networking Index published in 2019, it is predicted that "IoT connections will represent more than half (14.6 billion) of all global connected devices and connections (28.5 billion) by 2022." eMBB will be needed to handle this massive amount of traffic.

Massive Machine-Type Communications

Standards set by ITU for 5G networks mandate they be able to handle 1 million devices per square kilometer . This is especially needed for factories using Industrial IoT (IIoT) devices where there is a relatively low volume of non-delay-sensitive data, such as sensors. In a 5G network supporting an IIoT deployment, a single base station can be used to serve the deployment , thanks to the mMTC standard. Data packet sizes are small, which reduces congestion when there is a large amount of data from a large number of concurrent connections.

Ultra-Reliable and Low-Latency Communications

Mission-critical applications and IoT use cases that demand safety as a priority need highly-reliable connections with low latency. Those requirements necessitate the URLLC standard of 5G networks. For example, self-driving cars require low-latency. An autonomous car doesn't need a constant internet connection to run. However, when it needs to communicate with other cars or edge computing installations, low latency is mandatory so the car can make a decision before it hits another car, pedestrian, or other object. IoT medical devices need both reliable and low-latency connections. In the case of remote surgery, the connection between the devices on-site and the remote surgeon must be below 10 milliseconds . A bad connection or a slow one can have drastic life-or-death consequences. 5G networks have the potential to provide that with URLLC where latency is as low as 1 millisecond in ITU standards .



This diagram shows the primary concern for each mobile broadband (MBB) usage scenario. Source: Springer

IoT Networks: 4G LTE

The internet of things is not waiting patiently for 5G networks to mature. According to Cisco, IoT devices have been around since between 2008 and

2009 , long before 5G network development. For machine to machine (M2M) connections, LTE networks use the Cat-M1 chipset. M2M connections are the basis of mMTC. Narrowband IoT (NB-IoT) is a type of low power wide area (LPWA) technology that supports IoT devices. NB-IoT improves device power consumption, the capacity of the network system, and how efficiently the available spectrum is used.

The inclusion of Cat-M1 and NB-IoT means LTE networks can work with 5G network use cases, such as M2M communications. However, the number of devices served in a given geographic area is much less than in 5G networks. LTE networks can provide coverage of 60,680 LPWA devices per square kilometer . Compare this number to the one given above in the mMTC usage scenario.

According to Qualcomm , Cat-M1 chips and NB-IoT technology will be adopted by 5G networks for in-band deployments to connect low-complexity IoT devices. The standards organization, 3GPP, said in its Release 13 standards how mMTC and NB-IoT can work with both LTE networks and the 5G New Radio (NR) technology.

IoT Network

1. 5G standards have set expectations for the network to handle tens of billions of IoT network connections, 1 million connections per square kilometer, and scenarios where IoT devices need a reliable connection with low latency.
2. 4G LTE networks have long supported IoT devices with technology like the Cat-M1 chipset.
3. The growth of IoT devices and connections will eventually outstrip the capacity of LTE networks and will require the capabilities of mature 5G networks.

Components of an IoT

Here are the main components based on which an internet of things ecosystem works on.

1. Sensing and embedding components

This is the first tier of an IoT ecosystem and it forms the backbone of the entire Internet of Things network. Data is indispensable for IoT and sensors are an important factor to ensure the accuracy and credibility of data. This essential layer consists of physical, micro appliances, embedded in an IoT device , which are responsible for collecting data or controlling a mechanism.

Sensors

Sensors work to gather minute data from the surrounding environment. They are sometimes also known as 'detectors' as the primary function of sensors is to detect even the slightest changes in the surrounding environment. This allows an IoT device to capture relevant data for real-time or post-processing.

Depending on the type of sensor, this small piece of hardware can measure absolutely anything. This can be smoke, motion or even blood pressure. While advanced sensors can measure a range of complexities, some IoT devices have multiple sensors bundled to be able to collect a range of data or perform multiple functions. Our smartphones for example have GPS, fingerprint, camera, tilt, motion and numerous other sensors, all bundled in one.

Smart ACs or thermostats are able to sense room temperature and humidity levels at the same time. Depending on the device and use-case, different applications require different types of sensors.

Sensors are integral to achieve automation based on certain triggers. Considering the example of smart ACs, a person utilizing an automated mode function may set room temperature preferences between 73 and 77 degrees Fahrenheit. As soon as a room temperature higher than 77 degrees is detected the device will transmit a command to the air conditioning unit to operate at specified settings. As soon as a room is cooler than 73 degrees the change in temperature will be detected and a signal will be transmitted to the AC to turn off.

In the image below, an IoT-enabled smart AC controller is used to make a conventional air conditioner smart. It consists of a sensor which detects room temperature conditions along with a transmitter to send signals and receive a response. The entire IOT ecosystem is in play to guide automated actions.



Thanks to advancements in technology, today's sensors are minute, smart and cheap! The selection of sensors depends on the purpose you wish to achieve. You may want the sensor to be able to detect motion, temperature, pressure, smoke or any other such trigger. The choice of sensors also depends on their accuracy, reliability of results, the range at which they should work, resolution and level of intelligence which in other words means their ability to deal with noise and interference.

Actuators

Actuators work opposite to that of sensors. While sensors, sense; actuators act. They receive a signal or a command and on its basis they cause an action. They are as crucial as sensors as once the sensors have detected a change in the environment, an actuator is required to make something happen based on the trigger.

As an example, actuators may control the heating and cooling in a smart air conditioner or the valve in a smart tap. As soon as sensors detect that a person has exited an area, the actuator will be triggered to stop the flow of air conditioning or the flow of water in the case of a tap.

There are multiple types of actuators depending on the vertical and usage. They may be required to turn something on or off but they can also control valves and perform actions such as turning or gripping which has great beneficial value in industrial applications.

2. Connectivity

IoT is a network involving devices, sensors, cloud and actuators and all these need to interconnect with one another to be able to decipher data and consequently perform an action. Connectivity forms the second piece of the puzzle in the complex world of the IoT ecosystem.

Protocols

Once the data has been collected by the sensors, it requires a medium for transport. In other words, a communication channel is necessary between sensors and the cloud. IoT protocols are responsible for transferring data in the online world and this transmission can only be possible if two devices are safely connected. IoT standards and protocols involve an invisible language allowing physical objects to communicate with one another.

The choice of network depends on factors such as power consumption, speed of data transfer, range, bandwidth and overall efficiency. Some of the most popular IoT wireless protocols and standards include Bluetooth, Wi-Fi, ZigBee, LoRaWAN, DDS, MQTT, cellular, etc. These and other channels make it easy as well as secure to transfer and exchange data to the next IoT layer for processing.

IoT gateways

Incoming, raw data from the sensors must pass through gateways to reach the cloud. Gateways translate network protocols ensuring seamless communication of all devices within the network. Essentially this makes the gateways a crucial communication point and is responsible for easy management of data traffic.

Moreover, gateways offer security by protecting the system from unauthorized access and malicious attacks. It can also be considered as a security layer as the data flowing through it is protected by the latest encryption practices.

Gateways can also preprocess data from the sensors before sending it to the cloud. In other words they minimize the large volumes of data 'sensed' in the previous stage. Not all, but some intelligent IoT gateways have the ability to also analyze and average data to transfer only the relevant data to the cloud.

3. IoT cloud

Once the data has been collected and it has traveled to the cloud, it needs to be processed. The cloud is where the “smart stuff” takes place! This high-performance facility majorly ties the components to the IoT ecosystem together. It handles the data, stores it and makes decisions to make or break a deal. All of this is performed for colossal amounts of data in just under milliseconds – the time is critical for IoT, as especially in critical concerns such as health and safety, latency cannot be compromised.

While the main purpose of IoT solutions is to provide and act on real-time information, there needs to be a component that is able to handle enormous amounts of data to cater to the time-sensitive nature of the IoT model. This is where cloud systems come into play. They form the brain of the IoT ecosystem as they are typically responsible for processing, commanding or taking analytics into account for the collected data. Devices, protocols, gateway and storage are combined for efficient real-time data analysis.

With their immense computing power, storage capabilities, networking options, analytics and other service components, clouds make information effectively available for the consumers.

While the cloud is not necessary for IoT, since local processing with Edge or Fog computing is an option too, the cloud may be preferred being a high-performance facility that offers massive scalability and decreased operational costs. Edge computing on the other hand is preferred when large amounts of data processing and storage are required on-premises.

4. IoT analytics and data management

Data may be a small word but it holds immense power that can pose a huge effect on any business. IoT Analytics is used to make sense of the vast amounts of analog data. This for example can include the determination of key performance indicators in a certain application where one may be interested in viewing errors or irregularities in real-time.

Once identified an immediate action would be required to prevent any undesirable scenarios. To put it differently, analytics involves converting raw data into useful insights that later are interpreted or analyzed to drive decision making.

Smart analytics is useful in multiple scenarios. The basic role is to analyze a situation and formulate a decision based on this. This can be basic such

as analyzing if a room's temperature falls in an acceptable range, or complex if for example a car is just about to crash. Data analytics helps determine vital business insights. Deep learning models can be used for predictive analysis. Various learnings can be derived from the data to predict trends, plan ahead and make useful business decisions.

Analytics requires storage power and intelligent computation to be able to make sense of any data. Tasks such as this can be hosted on the cloud, depending on the IoT architecture.

5. End-user devices and user interface

The user interface is the visible component that is easily accessible and in control of the IoT user. This is where a user can control the system and set their preferences. The more user-friendly this component of the IoT ecosystem is, the easier is a user's interaction.

A user may interact with the system via the device itself, or this interaction can be conducted remotely via smartphones, tablets, and laptops. Smart home systems such as Amazon Alexa or Google Home etc. also allow users to communicate with their "things".

Design is a major consideration in today's fast-paced world and one IoT device can set itself apart from a competitor on the basis of a strong design. Touch interfaces, use of colors, font, voice, and more are some of the factors that come to play here. While an attractive design is necessary, the interface should be user-friendly enough to avoid any difficulties for the user.

REFERENCES:-

SUMMARY:-

Internet of Things refers to the rapidly growing network of connected objects that are able to collect and exchange data in real time using embedded sensors. Thermostats, cars, lights, refrigerators, and more appliances can all be connected to the IoT.

TOPIC:- IoT service oriented architecture, IoT challenges

OBJECTIVES:- SOA is an architectural style for building software applications that use services available in a network such as the web. It promotes loose coupling between software components so that they can be reused.

OUTCOME:- Service-Oriented Architecture makes it easier to deploy SaaS (software-as-a-service) in a cloud platform, while the cloud supplies the processing power and other resources required for the process. SOA and cloud integration makes it easier for your organization to collaborate within, and with IT as well.

IoT service oriented architecture, IoT challenges,

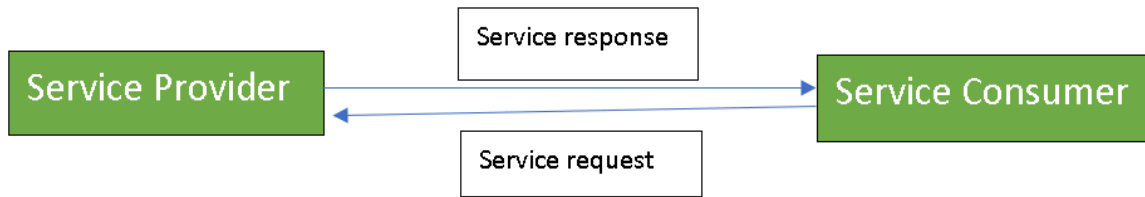
Service-Oriented Architecture (SOA) is an architectural approach in which applications make use of services available in the network. In this architecture, services are provided to form applications, through a communication call over the internet.

- SOA allows users to combine a large number of facilities from existing services to form applications.
- SOA encompasses a set of design principles that structure system development and provide means for integrating components into a coherent and decentralized system.
- SOA based computing packages functionalities into a set of interoperable services, which can be integrated into different software systems belonging to separate business domains.

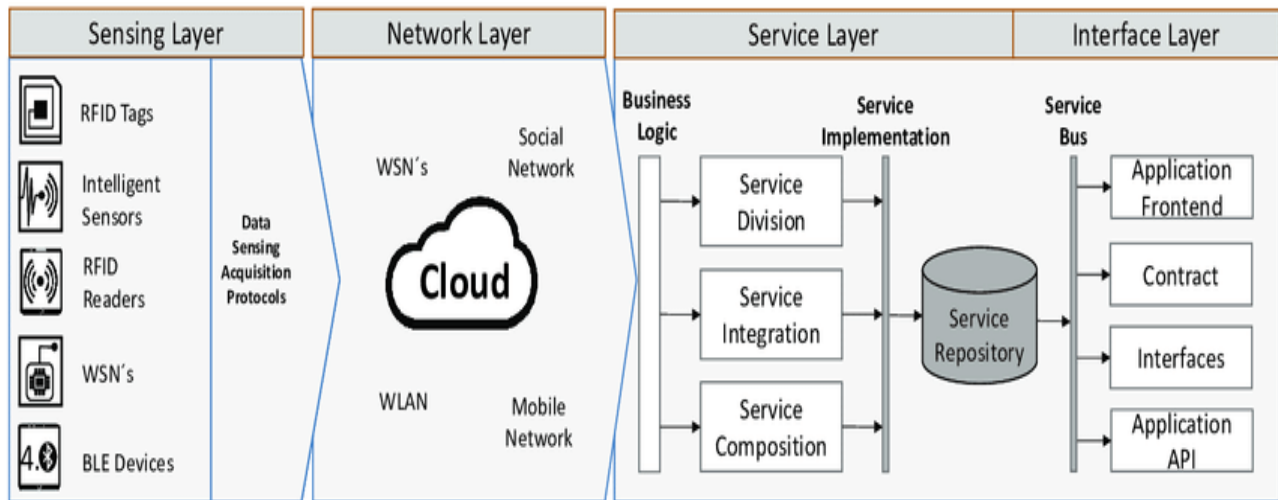
There are two major roles within Service-oriented Architecture:

1. **Service provider:** The service provider is the maintainer of the service and the organization that makes available one or more services for others to use. To advertise services, the provider can publish them in a registry, together with a service contract that specifies the nature of the service, how to use it, the requirements for the service, and the fees charged.
2. **Service consumer:** The service consumer can locate the service metadata in the registry and develop the required client components to

bind and use the service.



IoT service oriented architecture



Characteristics Of Service-Oriented Architecture

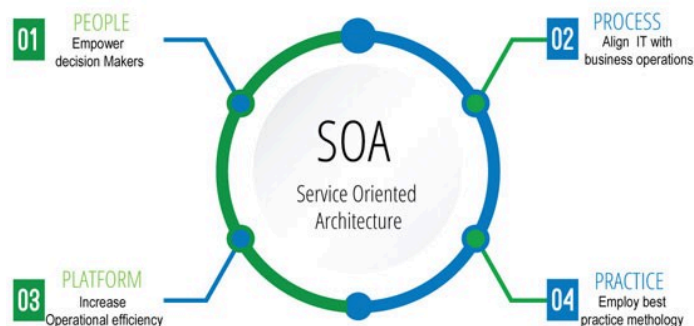
While the defining concepts of Service-Oriented Architecture vary from company to company, there are six key tenets that overarching the broad concept of Service-Oriented Architecture. These core values include:

- Business value
- Strategic goals
- Intrinsic inter-operability
- Shared services
- Flexibility
- Evolutionary refinement

Each of these core values can be seen on a continuum from older format distributed computing to Service-Oriented Architecture to cloud computing

(something that is often seen as an offshoot of Service-Oriented Architecture).

Service-Oriented Architecture Patterns



There are three roles in each of the Service-Oriented Architecture building blocks: service provider; service broker, service registry, service repository; and service requester/consumer.

The service provider works in conjunction with the service registry, debating the whys and hows of the services being offered, such as security, availability, what to charge, and more. This role also determines the service category and if there need to be any trading agreements.

The service broker makes information regarding the service available to those requesting it. The scope of the broker is determined by whoever implements it.

The service requester locates entries in the broker registry and then binds them to the service provider. They may or may not be able to access multiple services; that depends on the capability of the service requester.

Implementing Service-Oriented Architecture

When it comes to implementing service-oriented architecture (SOA), there is a wide range of technologies that can be used, depending on what your end goal is and what you're trying to accomplish.

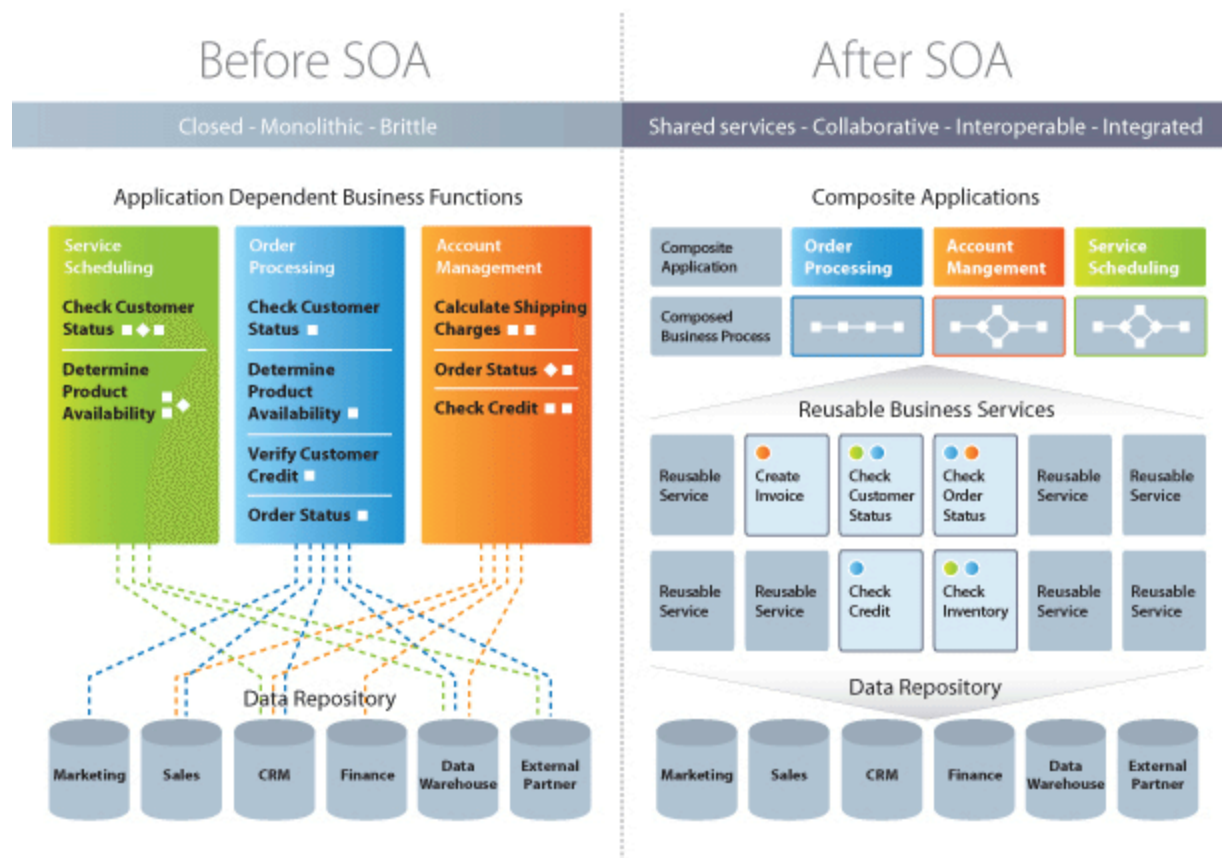
Typically, Service-Oriented Architecture is implemented with web services, which makes the "functional building blocks accessible over standard

internet protocols.”

An example of a web service standard is SOAP, which stands for Simple Object Access Protocol. In a nutshell, SOAP “is a messaging protocol specification for exchanging structured information in the implementation of web services in computer networks. Although SOAP wasn’t well-received at first, since 2003 it has gained more popularity and is becoming more widely used and accepted. Other options for implementing Service-Oriented Architecture include Jini, COBRA, or REST.

It’s important to note that architectures can “operate independently of specific technologies,” which means they can be implemented in a variety of ways, including messaging, such as ActiveMQ; Apache Thrift; and SORCER.

Service-Oriented Architecture Is Important



There are many benefits to service-oriented architecture , especially in a web service based business. We’ll outline a few of those benefits here, in brief:

Use Service-Oriented Architecture to create reusable code: Not only does this cut down on time spent on the development process, but there's no reason to reinvent the coding wheel every time you need to create a new service or process. Service-Oriented Architecture also allows for using multiple coding languages because everything runs through a central interface.

Use Service-Oriented Architecture to promote interaction: With Service-Oriented Architecture, a standard form of communication is put in place, allowing the various systems and platforms to function independent of each other. With this interaction, Service-Oriented Architecture is also able to work around firewalls, allowing "companies to share services that are vital to operations."

Use Service-Oriented Architecture for scalability: It's important to be able to scale a business to meet the needs of the client, however certain dependencies can get in the way of that scalability. Using Service-Oriented Architecture cuts back on the client-service interaction, which allows for greater scalability.

Use Service-Oriented Architecture to reduce costs: With Service-Oriented Architecture, it's possible to reduce costs while still "maintaining a desired level of output." Using Service-Oriented Architecture allows businesses to limit the amount of analysis required when developing custom solutions.

The Internet of Things (IoT) has quickly become a huge part of how people live, communicate and do business. All around the world, web-enabled devices are turning our world into a more switched-on place to live. There are many benefits to the increased adoption of IoT technology, says Kate Began, **Polycase** sales and marketing manager, from the ability to monitor cargo anywhere to playing your favorite music in the shower from a waterproof Bluetooth speaker. But there are still many challenges to widespread IoT adoption and to a secure, functioning global device network.

From security challenges to the perils of high customer expectations, these five factors are big concerns for the growth and development of the Internet of Things. Overcoming them will be the key to creating true lasting productivity and prosperity through these incredible technologies.

1. Security

Ask any security expert about the biggest headaches of the 21st century and they'll likely bring up IoT devices. The reason? In cybersecurity terms, IoT devices greatly expand the "attack surface," or the amount of potential areas for cybercriminals to penetrate a secure network.

Cybercriminals don't have to crack an IoT device's plastic enclosure to access sensitive materials. They can simply finesse their way in through one of the many security vulnerabilities that are found throughout the IoT. Many IoT devices have default passwords left unchanged, unpatched software and other major security vulnerabilities.

Much of the burden of fixing this problem falls upon IoT device users. Many people still don't see IoT devices as potential security threats that have to be patched, updated and protected in much the same way that smartphones and computers do.

(In fact, many people still don't protect their phones or computers well enough, either.) But, as we'll discuss below, governments often haven't moved with sufficient speed to regulate these new technologies as they become available.

2. Regulation

Another common characteristic of technological innovations is that government regulation often takes a long time to catch up with the current state of technology. With the rapid evolution that's happening every day in IoT, the government is taking its time in catching up and businesses are often left without crucial information they need to make decisions.

The lack of strong IoT regulations is a big part of why the IoT remains a severe security risk, and the problem is likely to get worse as the potential attack surface expands to include ever more crucial devices. When medical devices, cars and children's toys are all connected to the Internet, it's not hard to imagine many potential disaster scenarios unfolding in the absence of sufficient regulation.

Quality control in IoT can be particularly tricky from a regulatory perspective. With huge numbers of IoT devices now being imported from countries like China that have different standards of quality and security, many experts are calling for strong and universal security standards for IoT technology.

3. Compatibility

New waves of technology often feature a large stable of competitors jockeying for market share, and IoT is certainly no exception. This can be good news, since competition creates increased choices for consumers, but it can also create frustrating compatibility issues.

Home mesh networks are one area where compatibility trouble is looming.

Bluetooth has long been the compatibility standard for IoT devices. In fact, it was named after an ancient king, Harald Bluetooth, known for unifying warring tribes. But when it comes to home automation using mesh networking, several competitors have sprung up to challenge Bluetooth's mesh network offerings, including protocols such as Zigbee and Z-Wave. It could be years before the market settles enough to crown a single universal standard for home IoT.

Continued compatibility for IoT devices also depends upon users keeping their devices updated and patched, which, as we've just discussed, can be pretty difficult. When IoT devices that have to talk to each other are running different software versions, all kinds of performance issues and security vulnerabilities can result. That's a big part of why it's so important that IoT consumers keep their devices patched and up to date.

4. Bandwidth

Connectivity is a bigger challenge to the IoT than you might expect. As the size of the IoT market grows exponentially, some experts are concerned that bandwidth-intensive IoT applications such as video streaming will soon struggle for space on the IoT's current server-client model.

That's because the server-client model uses a centralised server to authenticate and direct traffic on IoT networks. However, as more and more devices begin to connect to these networks, they often struggle to bear the load.

Thus, it's important for IoT companies to carefully examine their IoT connectivity providers and to choose one with a strong record of service and innovation. Features like intelligent switching between mobile network operators (MNOs) are particularly useful for creating a more reliable and user-friendly IoT product for your customers.

5. Customer expectations

It's often said that it's better to under-promise and over-deliver. Many IoT manufacturers have learned this the hard way, with IoT start-ups failing often and leaving bewildered customers in their wake . When customer expectations and product reality don't match, the results can be system failures, orphaned technologies and lost productivity.

With such strong competition in the IoT market, customers whose expectations aren't met won't hesitate to go elsewhere. Businesses looking to enter this competitive and innovative sector should be prepared for a market that never sits still and customers who always want a smoother

and more advanced experience. IoT is an exciting sector with a lot of potential to change the way we live, work and play. But the tech industry, government and consumers alike must get on the same page about issues of security and performance to ensure that the IoT remains safe and productive to use.

REFERENCES:-

SUMMARY:-

1. Improves decision making processes, since executives could have more information available in less time.
2. Increases employee productivity.
3. Relationships with clients and suppliers are enhanced.

TOPIC:- 6LoWPAN, IEEE 802.15.4, ZigBee and its types

OBJECTIVES:- IEEE 802.15.4 is a technical standard which defines the operation of a low-rate wireless personal area network (LR-WPAN). It specifies the physical layer.

OUTCOME:-

6LoWPAN

It is an acronym of *IPv6 over Low -Power Wireless Personal Area Networks*. 6LoWPAN is the name of a concluded working group in the Internet area of the IETF.

The 6LoWPAN concept originated from the idea that "the Internet Protocol could and should be applied even to the smallest devices," and that low-power devices with limited processing capabilities should be able to participate in the Internet of Things.

The 6LoWPAN group has defined encapsulation and header compression mechanisms that allow IPv6 packets to be sent and received over IEEE 802.15.4 based networks. IPv4 and IPv6 are the work horses for data delivery for local-area networks, metropolitan area networks, and wide-area networks such as the Internet. Likewise, IEEE 802.15.4 devices provide sensing communication-ability in the wireless domain. The inherent natures of the two networks though, are different.

Application areas

The target for IP networking for low-power radio communication is applications that need wireless internet connectivity at lower data rates for devices with very limited form factor. An example is automation and entertainment applications in home, office and factory environments. The header compression mechanisms standardized in RFC6282 can be used to provide header compression of IPv6 packets over such networks.

IPv6 is also in use on the smart grid enabling smart meters and other devices to build a micro mesh network before sending the data back to the billing system using the IPv6 backbone. Some of these networks run over IEEE 802.15.4 radios, and therefore use the header compression and fragmentation as specified by RFC6282.

Thread

Thread is an effort of over 50 companies to standardize on a protocol running over 6LoWPAN to enable home automation. The specification is

available at no cost as of 7 May 2018, subject to adherence to an EULA stipulating that Thread Group membership (in most cases paid) is required to implement the protocol.^[5] ^[6] It is to be launched in the second half of 2018. The protocol will most directly compete with Z-Wave and Zigbee IP.

Connected Home over IP

Connected Home over IP is the latest effort to standardize on a protocol running over 6LoWPAN to enable home automation, by combining it with DTLS, CoAP and MQTT-SN.

Functions

As with all link-layer mappings of IP, RFC4944 provides a number of functions. Beyond the usual differences between L2 and L3 networks, mapping from the IPv6 network to the IEEE 802.15.4 network poses additional design challenges.

Adapting the packet sizes of the two networks

IPv6 requires the maximum transmission unit (MTU) to be at least 1280 octets. In contrast, IEEE 802.15.4's standard packet size is 127 octets. A maximum frame overhead of 25 octets spares 102 octets at the media access control layer. An optional but highly recommended security feature at the link layer poses an additional overhead. For example, 21 octets are consumed for AES-CCM-128 leaving only 81 octets for upper layers.

The concept was created because engineers felt like the smallest devices were being left out from the Internet of Things. 6LoWPAN can communicate with 802.15.4 devices as well as other types of devices on an IP network link like WiFi. A bridge device can connect the two.

6LoWPAN's M2M/IoT Applications

- 6LoWPan Smart Meters
- Smart Home (Lighting, Thermostats)

Basically anything that is relatively low-power, but can operate in close proximity to its neighbor transceivers.

ZigBee

ZigBee, like 6LoWPAN, is designed for low data-rate and battery-powered applications. ZigBee is the most popular, low-cost, low-power wireless mesh networking standard on the market right now—and the more mature

technology of the two (ZigBee, 6LoWPAN). It is typically implemented for personal or home-area networks, or in a wireless mesh for networks that operate over longer ranges.

The ZigBee IP is built on the IEEE 802.15.4 standard, but unlike 6LoWPAN, it cannot easily communicate with other protocols. A benefit of ZigBee, however, is that nodes can stay in sleep mode most of the time, drastically extending battery life. There is a new type of Zigbee-like mesh technology called Z-Wave also, read more about Z-Wave vs Zigbee [here](#).

ZigBee's M2M/IoT Applications

- Wireless Light Switches
- Electrical Meters (Smart grid, demand response, etc)
- Industrial Equipment Monitoring

The name is based on the wiggly dance that honey bees do on their way to drop off their honey. Like those crazy bees, data in a ZigBee network is "hopping" around a mesh of transceivers until a route to the host (usually the internet) is found. It is based on the IEEE 802.15 protocol, and has a fixed data-rate of 250 kbit/s. This speed, coupled with low-transmit power, means ZigBee does not have great range. Repeaters and/or a high density of nodes are often needed to obtain the desired coverage.

IEEE 802.15.4

IEEE 802.15.4 is a technical standard which defines the operation of low-rate wireless personal area networks (LR-WPANs). It specifies the physical layer and media access control for LR-WPANs, and is maintained by the IEEE 802.15 working group, which defined the standard in 2003.^[1] It is the basis for the Zigbee^[2], ISA100.11a^[3],

WirelessHART, MiWi, 6LoWPAN, Thread and SNAP specifications, each of which further extends the standard by developing the upper layers which are not defined in IEEE 802.15.4. In particular, 6LoWPAN defines a binding for the IPv6 version of the Internet Protocol (IP) over WPANs, and is itself used by upper layers like Thread.

IEEE standard 802.15.4 intends to offer the fundamental lower network layers of a type of wireless personal area network (WPAN) which focuses on low-cost, low-speed ubiquitous communication between devices. It can be contrasted with other approaches, such as Wi-Fi, which offer more bandwidth and requires more power. The emphasis is on very low cost communication of nearby devices with little to no underlying infrastructure, intending to exploit this to lower power consumption even more.

IEEE 802.15.4 is a low-data rate wireless personal area network and is the PHY and MAC layer used by many IoT protocols, such as ZigBee, and WirelessHART.

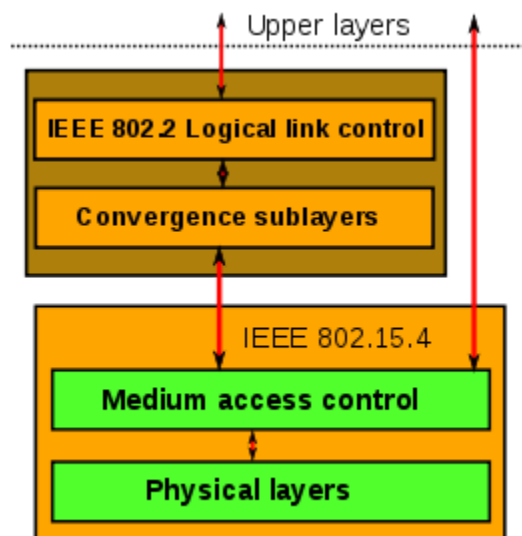
The basic framework conceives a 10-meter communications range with a transfer rate of 250 kbit/s. Tradeoffs are possible to favor more radically embedded devices with even lower power requirements, through the definition of not one, but several physical layers. Lower transfer rates of 20 and 40 kbit/s were initially defined, with the 100 kbit/s rate being added in the current revision.

Even lower rates can be used, which results in lower power consumption. As already mentioned, the main goal of IEEE 802.15.4 regarding WPANs is the emphasis on achieving low manufacturing and operating costs through the use of relatively simple transceivers, while enabling application flexibility and adaptability.

Key 802.15.4 features include

1. real-time suitability by reservation of Guaranteed Time Slots (GTS),
2. collision avoidance through CSMA/CA
3. integrated support for secure communications.
4. power management functions such as link speed/quality and energy detection.
5. Support for time and data rate sensitive applications because of its ability to operate either as CSMA/CA or TDMA access modes. The TDMA mode of operation is supported via the GTS feature of the standard.^[4]
6. IEEE 802.15.4-conformant devices may use one of three possible frequency bands for operation (868/915/2450 MHz).

Protocol architecture



IEEE 802.15.4 protocol stack

Devices are designed to interact with each other over a conceptually simple wireless network . The definition of the network layers is based on the OSI model ; although only the lower layers are defined in the standard, interaction with upper layers is intended, possibly using an IEEE 802.2 logical link control sublayer accessing the MAC through a convergence sublayer. Implementations may rely on external devices or be purely embedded, self-functioning devices.

Zigbee is an IEEE 802.15.4 -based specification for a suite of high-level communication protocols used to create personal area networks with small, low-power digital radios , such as for home automation , medical device data collection, and other low-power low-bandwidth needs, designed for small scale projects which need wireless connection. Hence, Zigbee is a low-power, low data rate, and close proximity (i.e., personal area) wireless ad hoc network .

The technology defined by the Zigbee specification is intended to be simpler and less expensive than other wireless personal area networks (WPANs), such as Bluetooth or more general wireless networking such as Wi-Fi . Applications include wireless light switches, home energy monitors , traffic management systems, and other consumer and industrial equipment that requires short-range low-rate wireless data transfer.

Its low power consumption limits transmission distances to

10–100 meters line-of-sight , depending on power output and environmental characteristics.^[2] Zigbee devices can transmit data over long distances by passing data through a mesh network of intermediate devices to reach more distant ones. Zigbee is typically used in low data rate applications that require long battery life and secure networking (Zigbee networks are secured by 128 bit symmetric encryption keys.) Zigbee has a defined rate of 250 kbit/s, best suited for intermittent data transmissions from a sensor or input device.

REFERENCES:-

SUMMARY:-

Devices are designed to interact with each other over a conceptually simple wireless network

TOPIC:- RFID Features, RFID working principle and applications

OBJECTIVES:- Its is a technology based identification system which helps identifying objects

OUTCOME:-

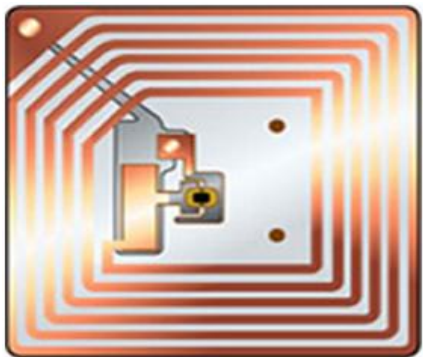
Defining RFID:

RFID or Radio Frequency Identification System is a technology based identification system which helps identifying objects just through the tags attached to them, without requiring any light of sight between the tags and the tag reader. All that is needed is radio communication between the tag and the reader.

A Basic RFID System:

3 Main Components of a RFID System

A RFID tag: It consists of a silicon microchip attached to a small antenna and mounted on a substrate and encapsulated in different materials like plastic or glass veil and with an adhesive on the back side to be attached to objects.



RFID Tag

- **A reader:** It consists of a scanner with antennas to transmit and receive signals and is responsible for communication with the tag and receives the information from the tag.



An RFID Reader

- **A Processor or a Controller:** It can be a host computer with a Microprocessor or a microcontroller which receives the reader input and process the data.

2 Types of RFID Systems:

- **Active RFID system:** These are systems where the tag has its own power source like any external power supply unit or a battery. The only constraint being the life time of the power devices. These systems can be used for larger distances and to track high value goods like vehicles.
- **Passive RFID system:** These are systems where the tag gets power through the transfer of power from a reader antenna to the tag antenna. They are used for short range transmission.

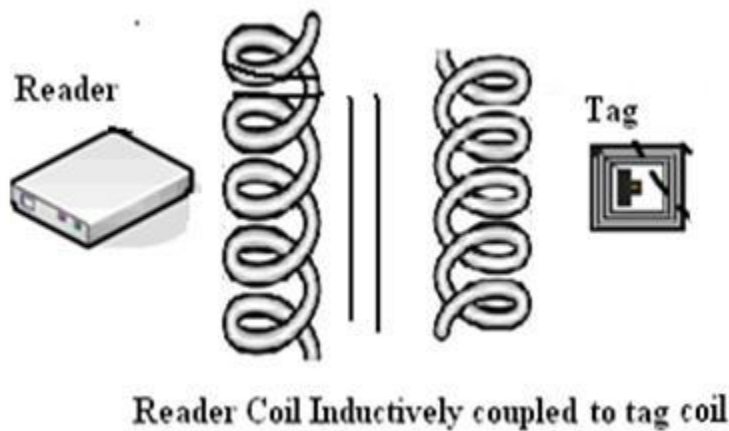
Here we are mostly concerned with the passive RFID system as it is most widely used in regular applications like in retail market organizations.

A brief idea about How the Passive RFID System Works:

The tag can be powered either using inducting coupling method or through EM wave capture method. Lets us have a brief knowledge about the system using these two methods.

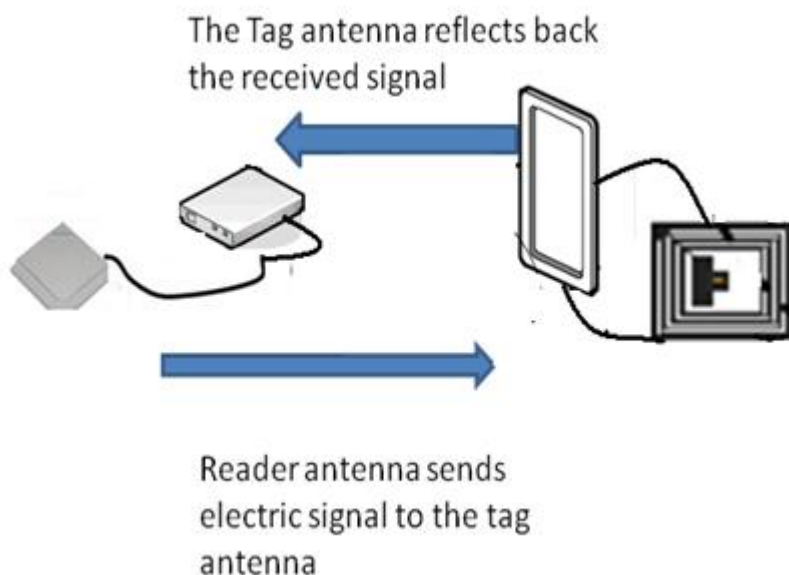
- **A Passive RFID system using Induction coupling method:** In this approach the RFID tag gets power from the reader through inductive coupling method. The reader consists of a coil connected to an AC supply such that a magnetic field is formed around it. The tag coil is placed in the vicinity of the reader coil and an electromotive force is induced it by the virtue of Faraday's law of induction. The EMF causes a flow of current in the coil, thus producing a magnetic field around it. By the virtue of Lenz law, the magnetic field of the tag coil opposes the reader's magnetic field and there will be a subsequent increase in the

current through the reader coil. The reader intercepts this as the load information. This system is suitable for very short distance communication. The AC voltage appearing across the tag coil is converted to DC using rectifier and filter arrangement.



Passive RFID using Inductive Coupling

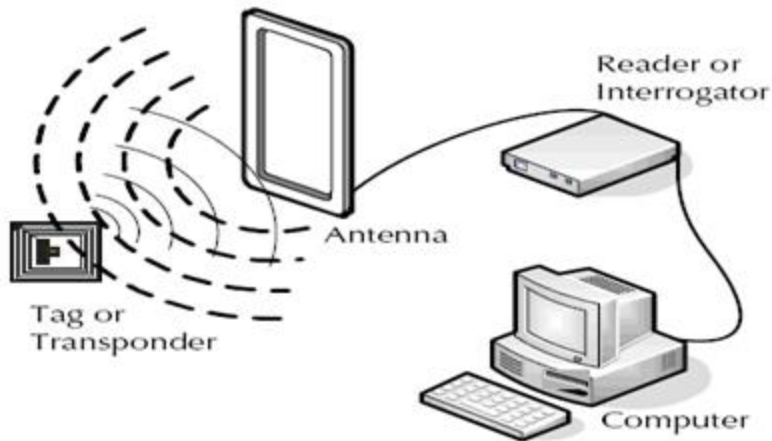
- **A Passive RFID system using EM wave propagation method:** The antenna present in the reader transmits electromagnetic waves which are received by the antenna present in the tag as potential difference across the dipole. This voltage is rectified and filtered to get the DC power. The receiver antenna is kept at different impedance which causes it to reflect a part of the received signal. This reflected signal is received by the reader and monitored accordingly.



Passive RFID using EM-wave transmission

An Idea about How the Active RFID System Works:

In the active RFID system, the reader sends signal to the tag using an antenna. The tag receives this information and resends this information along with the information in its memory. The reader receives this signal and transmits to the processor for further processing.

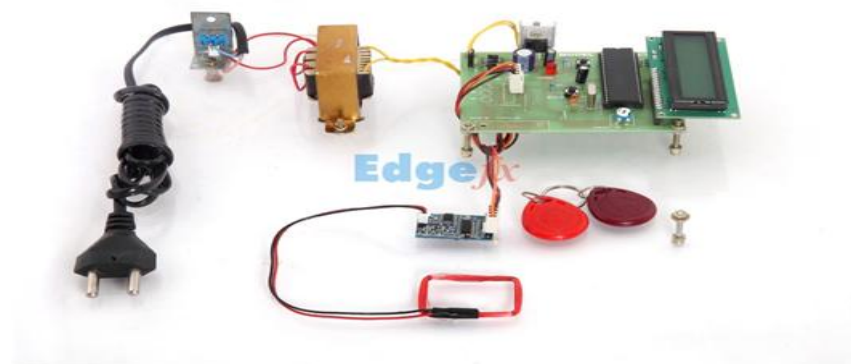


An Active RFID system

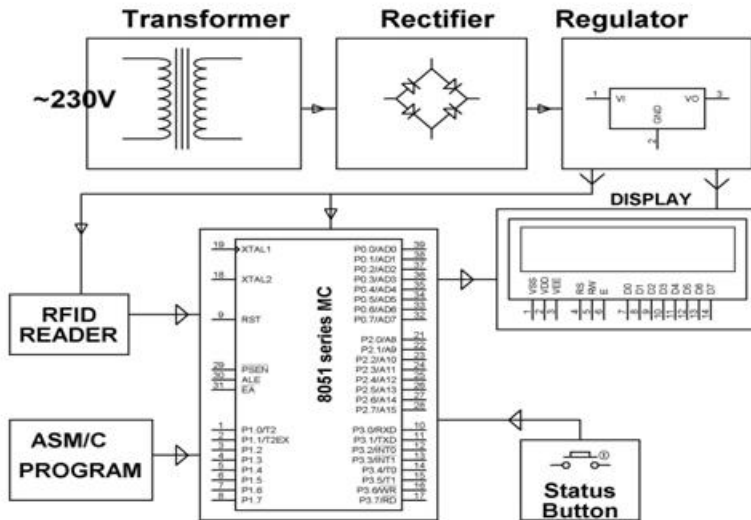
A Working Example of RFID Application – An RFID Based Attendance System

So, now let us see a practical approach to the solution of our second problem – Maintaining and checking a database for members of an institution, using RFID system.

The basic idea involves each person of the institution having an id card and when this card is swiped against the reader, the person's info is matched with the existing system in the database and the his/her attendance is marked.



A Practical RFID based Attendance System



Block Diagram of the System

The whole system uses the passive RFID system with inductive coupling method. As the RFID card (tag) is swiped against the RFID reader, a carrier signal of 125 KHz is sent to the tag coil, which receives this signal and modulates them. This modulated signal is received by the reader, interfaced to the microcontroller. The microcontroller receives this data and is programmed to compare it with the data in the existing database. If the data matches, the relevant details of that particular person is displayed on the LCD interfaced to the microcontroller.

REFERENCES:-

SUMMARY:-

TOPIC:- NFC (Near Field communication), Bluetooth

OBJECTIVES:-

OUTCOME:-

Near Field Communication (NFC)

Near Field Communication (NFC) is a short-range wireless connectivity standard that uses magnetic field induction to enable communication between devices when they're touched together, or brought within a few centimeters of each other. Jointly developed by Philips and Sony, the standard specifies a way for the devices to establish a peer-to-peer (P2P) network to exchange data. After the P2P network has been configured, another wireless communication technology, such as Bluetooth or Wi-Fi, can be used for longer range communication or for transferring larger amounts of data.

Here are some examples of how NFC can be used:

- You could take pictures with a cell phone with a built in camera, and touch an enabled computer or television set to transmit the images for display;
- You could download applications or games to a handheld device by touching the computer;
- In conjunction with another wireless technology, you could transfer large files between two devices, such as a laptop and a desktop, simply by touching the two together.

Every active NFC device can work in one or more of three modes:

NFC card emulation

Enables NFC-enabled devices such as smartphones to act like smart cards, allowing users to perform transactions such as payment or ticketing.

NFC reader/writer

Enables NFC-enabled devices to read information stored on inexpensive NFC tags embedded in labels or smart posters.

NFC peer-to-peer

Enables two NFC-enabled devices to communicate with each other to

exchange information in an ad hoc fashion.

- Standards
 - ISO / IEC
 - GSMA
 - StoLPaN
 - NFC Forum
 - Other standardization bodied

There are many other uses for NFC as well. Examples include:

- Paying a fare on public transit, such as a bus or train
- Confirming your ticket at a concert or sports event
- Syncing workout data from a fitness machine with your device
- Viewing special offers on your phone when you enter a store
- Loading information about an artist or piece of art at a museum
- Viewing a map and related information at a national park
- Loading a translated menu at a restaurant
- Checking in and checking out at a hotel
- Unlocking an NFC-enabled door lock.

Bluetooth

It is a wireless technology standard used for exchanging data between fixed and mobile devices over short distances using UHF radio waves in the ISM bands, from 2.402 GHz to 2.480 GHz, and building personal area networks (PANs). It was originally conceived as a wireless alternative to RS-232 data cables.

Bluetooth is a short-range wireless communication technology standard. Bluetooth is managed by the Bluetooth Special Interest Group (SIG), which has more than 35,000 member companies in the areas of telecommunication, computing, networking, and consumer electronics. The IEEE standardized Bluetooth as **IEEE 802.15.1**, but no longer maintains the standard. The Bluetooth SIG oversees development of the specification, manages the qualification program, and protects the trademarks. A manufacturer must meet Bluetooth SIG standards to market it as a Bluetooth device.^[4] A network of patents apply to the technology, which are licensed to individual qualifying devices. As of 2009, Bluetooth integrated circuit chips ship approximately 920 million units annually.

List of applications



A typical Bluetooth mobile phone headset

- Wireless control and communication between a mobile phone and a handsfree headset . This was one of the earliest applications to become popular.
- Wireless control of and communication between a mobile phone and a Bluetooth compatible car stereo system (and sometimes between the SIM card and the car phone).
- Wireless communication between a smartphone and a smart lock for unlocking doors.
- Wireless control of and communication with iOS and Android device phones, tablets and portable wireless speakers .
- Wireless Bluetooth headset and Intercom . Idiomatically, a headset is sometimes called "a Bluetooth".
- Wireless streaming of audio to headphones with or without communication capabilities.
- Wireless streaming of data collected by Bluetooth-enabled fitness devices to phone or PC.
- Wireless networking between PCs in a confined space and where little bandwidth is required.

REFERENCES:-

SUMMARY:-

TOPIC:- Wireless Sensor Networks and its Applications

OBJECTIVES:- A Wireless Sensor Network is one kind of wireless network that includes a large number of circulating, self-directed, minute, low powered devices named sensor nodes called motes.

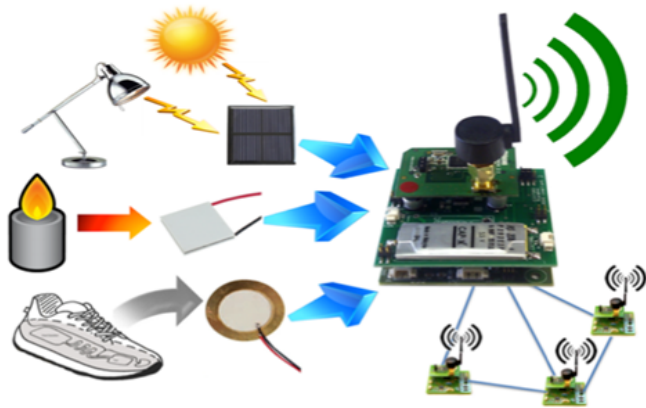
OUTCOME:-

Wireless Sensor Networks and its Applications

WSN (Wireless Sensor Network) is the most standard services employed in commercial and industrial applications, because of its technical development in a processor, communication, and low-power usage of embedded computing devices. The wireless sensor network architecture is built with nodes that are used to observe the surroundings like temperature, humidity, pressure, position, vibration, sound, etc. These nodes can be used in various real-time applications to perform various tasks like smart detecting, a discovery of neighbor nodes, data processing and storage, data collection, target tracking, monitor and controlling, synchronization, node localization, and effective routing between the base station and nodes. Presently, WSNs are beginning to be organized in an enhanced step. It is not awkward to expect that in 10 to 15 years that the world will be protected with WSNs with entree to them via the Internet. This can be measured as the Internet becoming a physical n/w. This technology is thrilling with infinite potential for many application areas like medical, environmental, transportation, military, entertainment, homeland defense, crisis management, and also smart spaces.

Wireless Sensor Network

A Wireless Sensor Network is one kind of wireless network that includes a large number of circulating, self-directed, minute, low powered devices named sensor nodes called motes. These networks certainly cover a huge number of spatially distributed, little, battery-operated, embedded devices that are networked to caringly collect, process, and transfer data to the operators, and it has controlled the capabilities of computing & processing. Nodes are tiny computers, which work jointly to form networks.



Wireless Sensor Network

The sensor node is a multi-functional, energy-efficient wireless device. The applications of motes in industrial are widespread. A collection of sensor nodes collects the data from the surroundings to achieve specific application objectives. The communication between motes can be done with each other using transceivers. In a wireless sensor network, the number of motes can be in the order of hundreds/ even thousands. In contrast with sensor n/ws, Ad Hoc networks will have fewer nodes without any structure.

Wireless Sensor Network Architecture

The most common wireless sensor network architecture follows the OSI architecture Model. The architecture of the WSN includes five layers and three cross layers. Mostly in sensor n/w, we require five layers, namely application, transport, n/w, data link & physical layer. The three cross planes are namely power management, mobility management, and task management. These layers of the WSN are used to accomplish the n/w and make the sensors work together in order to raise the complete efficiency of the network. Please follow the below link for Types of wireless sensor networks and WSN topologies

Types of WSN Architectures

The architecture used in WSN is sensor network architecture. This kind of architecture is applicable in different places such as hospitals, schools, roads, buildings as well as it is used in different applications such as security management, disaster management & crisis management, etc. There are two types of architectures used in wireless sensor networks which include the following. There are 2 types of wireless sensor architectures: Layered Network Architecture, and Clustered Architecture. These are explained as following below.

- Layered Network Architecture
- Clustered Network Architecture

Layered Network Architecture

This kind of network uses hundreds of sensor nodes as well as a base station. Here the arrangement of network nodes can be done into concentric layers. It comprises five layers as well as 3 cross layers which include the following.

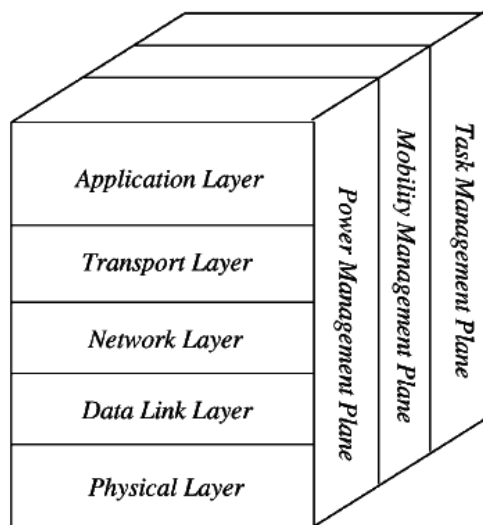
The five layers in the architecture are:

- Application Layer
- Transport Layer
- Network Layer
- Data Link Layer
- Physical Layer

The three cross layers include the following:

- Power Management Plane
- Mobility Management Plane
- Task Management Plane

These three cross layers are mainly used for controlling the network as well as to make the sensors function as one in order to enhance the overall network efficiency. The above mentioned five layers of WSN are discussed below.



Wireless Sensor Network Architecture

Application Layer

The application layer is liable for traffic management and offers software for numerous applications that convert the data in a clear form to find positive information. Sensor networks arranged in numerous applications in different fields such as agricultural, military, environment, medical, etc.

Transport Layer

The function of the transport layer is to deliver congestion avoidance and reliability where a lot of protocols intended to offer this function are either practical on the upstream. These protocols use dissimilar mechanisms for loss recognition and loss recovery. The transport layer is exactly needed when a system is planned to contact other networks.

Providing a reliable loss recovery is more energy-efficient and that is one of the main reasons why TCP is not fit for WSN. In general, Transport layers can be separated into Packet driven, Event-driven. There are some popular protocols in the transport layer namely STCP (Sensor Transmission Control Protocol), PORT (Price-Oriented Reliable Transport Protocol and PSFQ (pump slow fetch quick).

Network Layer

The main function of the network layer is routing, it has a lot of tasks based on the application, but actually, the main tasks are in the power conserving, partial memory, buffers, and sensor don't have a universal ID and have to be self-organized.

The simple idea of the routing protocol is to explain a reliable lane and redundant lanes, according to a convincing scale called a metric, which varies from protocol to protocol. There are a lot of existing protocols for this network layer, they can be separated into; flat routing and hierarchal routing or can be separated into time-driven, query-driven & event-driven.

Data Link Layer

The data link layer is liable for multiplexing data frame detection, data streams, MAC, & error control, confirm the reliability of point-point (or) point- multipoint.

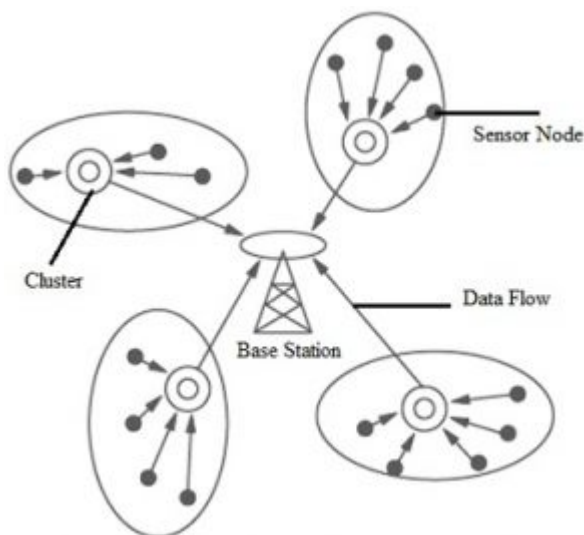
Physical Layer

The physical layer provides an edge for transferring a stream of bits above the physical medium. This layer is responsible for the selection of frequency, generation of a carrier frequency, signal detection, Modulation & data encryption. IEEE 802.15.4 is suggested as typical for low rate particular areas & wireless sensor networks with low cost, power consumption, density, the range of communication to improve the battery life. CSMA/CA is used to support star & peer to peer topology. There are several versions of IEEE 802.15.4.V.

The main benefits of using this kind of architecture in WSN is that every node involves simply in less-distance, low- power transmissions to the neighboring nodes due to which power utilization is low as compared with other kinds of sensor network architecture. This kind of network is scalable as well as includes a high fault tolerance.

Clustered Network Architecture

In this kind of architecture, separately sensor nodes add into groups known as clusters which depend on the “Leach Protocol” because it uses clusters. The term ‘Leach Protocol’ stands for “Low Energy Adaptive Clustering Hierarchy”. The main properties of this protocol mainly include the following.



Clustered Network Architecture

- This is a two-tier hierarchy clustering architecture.
- This distributed algorithm is used to arrange the sensor nodes into

groups, known as clusters.

- In every cluster which is formed separately, the head nodes of the cluster will create the TDMA (Time-division multiple access) plans.
- It uses the Data Fusion concept so that it will make the network energy efficient.

This kind of network architecture is extremely used due to the data fusion property. In every cluster, every node can interact through the head of the cluster to get the data. All the clusters will share their collected data toward the base station. The formation of a cluster, as well as its head selection in each cluster, is an independent as well as autonomous distributed method.

Design Issues of Wireless Sensor Network Architecture

The design issues of wireless sensor network architecture mainly include the following.

- Energy Consumption
- Localization
- Coverage
- Clocks
- Computation
- Cost of Production
- Design of Hardware
- Quality of Service

Energy Consumption

In WSN, power consumption is one of the main issues. As an energy source, the battery is used by equipping with sensor nodes. The sensor network is arranged within dangerous situations so it turns complicated for changing otherwise recharging batteries. The energy consumption mainly depends on the sensor nodes' operations like communication, sensing & data processing. Throughout communication, the energy consumption is very high. So, energy consumption can be avoided at every layer by using efficient routing protocols.

Localization

For the operation of the network, the basic, as well as critical problem, is sensor localization. So sensor nodes are arranged in an ad-hoc manner so they don't know about their location. The difficulty of determining the sensor's physical location once they have been arranged is known as localization. This difficulty can be resolved through GPS, beacon nodes,

localization based on proximity.

Coverage

The sensor nodes in the wireless sensor network utilize a coverage algorithm for detecting data as well as transmit them to sink through the routing algorithm. To cover the whole network, the sensor nodes should be chosen. There efficient methods like least and highest exposure path algorithms as well as coverage design protocol are recommended.

Clocks

In WSN, clock synchronization is a serious service. The main function of this synchronization is to offer an ordinary timescale for the nodes of local clocks within sensor networks. These clocks must be synchronized within some applications like monitoring as well as tracking.

Computation

The computation can be defined as the sum of data that continues through each node. The main issue within computation is that it must reduce the utilization of resources. If the life span of the base station is more dangerous, then data processing will be completed at each node before data transmitting toward the base station. At every node, if we have some resources then the whole computation should be done at the sink.

Production Cost

In WSN, the large number of sensor nodes is arranged. So if the single node price is very high then the overall network price will also be high. Ultimately, the price of each sensor node has to be kept less. So the price of every sensor node within the wireless sensor network is a demanding problem.

Hardware Design

When designing any sensor network's hardware like power control, micro-controller & communication unit must be energy-efficient. Its design can be done in such a way that it uses low-energy.

Quality of Service

The quality of service or QoS is nothing but, the data must be distributed in time. Because some of the real-time sensor-based applications mainly depend on time. So if the data is not distributed on time toward the receiver then the data will turn useless. In WSNs, there are different types of QoS

issues like network topology that may modify frequently as well as the accessible state of information used for routing can be imprecise.

Structure of a Wireless Sensor Network

The structure of WSN mainly comprises various topologies used for radio communications networks like a star, mesh, and hybrid star. These topologies are discussed below in brief.

Star Network

The communication topology like a star network is used wherever only the base station can transmit or receive a message toward remote nodes. There is a number of nodes are available which are not allowed to transmit messages to each other. The benefits of this network mainly comprise simplicity, capable of keeping the power utilization of remote nodes to a minimum.

It also lets communications with less latency among the base station as well as a remote node. The main drawback of this network is that the base station should be in the range of radio for all the separate nodes. It is not robust like other networks because it depends on a single node to handle the network.

Mesh Network

This kind of network permits to the transmission of the data from one node to another within the network that is in the range of radio transmission. If a node needs to transmit a message to another node and that is out of radio communications range, then it can utilize a node like an intermediate to send the message toward the preferred node.

The main benefit of a mesh network is scalability as well as redundancy. When an individual node stops working, a remote node can converse to any other type of node within the range, then forwards the message toward the preferred location. Additionally, the network range is not automatically restricted through the range among single nodes; it can extend simply by adding a number of nodes to the system.

The main drawback of this kind of network is power utilization for the network nodes that execute the communications like multi-hop are usually higher than other nodes that don't have this capacity of limiting the life of battery frequently. Moreover, when the number of communication hops increases toward a destination, then the time taken to send the message will also increase, particularly if the low power process of the nodes is a

necessity.

Hybrid Star – Mesh Network

A hybrid among the two networks like star and mesh provides a strong and flexible communications network while maintaining the power consumption of wireless sensor nodes to a minimum. In this kind of network topology, the sensor nodes with less power are not allowed to transmit the messages.

This permits to maintenance least power utilization.

But, other network nodes are allowed with the capability of multi-hop by allowing them to transmit messages from one node to another on the network. Usually, the nodes with the multi-hop capacity have high power and are frequently plugged into the mains line. This is the implemented topology through the upcoming standard mesh networking called ZigBee.

Structure of a Wireless Sensor Node

The components used to make a wireless sensor node are different units like sensing, processing, transceiver & power. It also includes additional components that depend on an application like a power generator, a location finding system & a mobilizer. Generally, sensing units include two subunits namely ADCs as well as sensors. Here sensors generate analog signals which can be changed to digital signals with the help of ADC, after that it transmits to the processing unit.

Generally, this unit can be associated through a tiny storage unit to handle the actions to make the sensor node work with the other nodes to achieve the allocated sensing tasks. The sensor node can be connected to the network with the help of a transceiver unit. In the sensor node, one of the essential components is a sensor node. The power-units are supported through power scavenge units like solar cells whereas the other subunits depend on the application.

A wireless sensing nodes functional block diagram is shown above. These modules give a versatile platform to deal with the requirements of wide applications. For instance, based on the sensors to be arranged, the replacement of signal conditioning block can be done. This permits to use of different sensors along with the wireless sensing node. Likewise, the radio link can be exchanged for a specified application.

Characteristics of Wireless Sensor Network

The characteristics of WSN include the following.

- The consumption of Power limits for nodes with batteries
- Capacity to handle node failures
- Some mobility of nodes and Heterogeneity of nodes
- Scalability to a large scale of distribution
- Capability to ensure strict environmental conditions
- Simple to use
- Cross-layer design

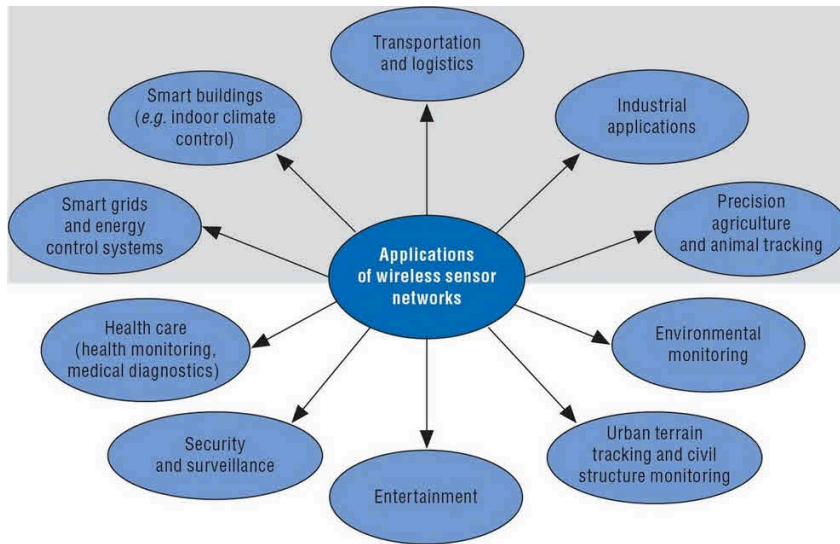
Advantages of Wireless Sensor Networks

The advantages of WSN include the following

- Network arrangements can be carried out without immovable infrastructure.
- Apt for the non-reachable places like mountains, over the sea, rural areas, and deep forests.
- Flexible if there is a casual situation when an additional workstation is required.
- Execution pricing is inexpensive.
- It avoids plenty of wiring.
- It might provide accommodations for the new devices at any time.
- It can be opened by using centralized monitoring.

Wireless Sensor Network Applications

Wireless sensor networks may comprise numerous different types of sensors like low sampling rate, seismic, magnetic, thermal, visual, infrared, radar, and acoustic, which are clever to monitor a wide range of ambient situations. Sensor nodes are used for constant sensing, event ID, event detection & local control of actuators. The applications of wireless sensor networks mainly include health, military, environmental, home, & other commercial areas.



WSN Application

- Military Applications
- Health Applications
- Environmental Applications
- Home Applications
- Commercial Applications
- Area monitoring
- Health care monitoring
- Environmental/Earth sensings
- Air pollution monitoring
- Forest fire detection
- Landslide detection
- Water quality monitoring
- Industrial monitoring